Thermistor based digital thermometers - For Clinical applications

The Centre for Materials for Electronics Technology is a premier research centre in Electronic materials under Ministry of Electronics and Information Technology, Government of India. C-MET is engaged in the development of process/product/technology and transfer of the technology in the area of Electronic materials, components and devices to cater to India's strategic and industrial applications, exploiting indigenous resources of raw materials.

The C-MET family comprises three Centres, located at Pune, Hyderabad and Thrissur with specialized research mandate at each place. C-MET, Thrissur is concentrating on the development of electronic ceramic materials and devices. Its main areas of focus include nano-materials, nano-porous materials, sensors and actuators, microwave materials & nano-structured electronics.

Monitoring of body temperature has been an effective tool to diagnose the health condition since prehistoric period. With the advancements in modern medicine, measurement and control of body temperature has become a vital aspect in health management. The most commonly used temperature measuring device in health sector, till the last decade, was mercury thermometers. However, there has been a gradual increase in the adoption of mercury-free temperature monitoring devices worldwide over the past few years. Efforts of many organisations in spreading awareness about the harmful effects of mercury in the environment through various initiatives have fuelled the higher adoption of mercury-free temperature monitoring devices and hence clinical digital thermometers have taken up a unique spot in the thermometer market.

The market growth of clinical digital thermometer has been accelerated by various factors such as rising prevalence of infectious diseases, increasing geriatric and pediatric population, increasing number of surgical procedures, growing number of blood transfusion procedures, rising awareness about continuous temperature monitoring, and increasing demand for advanced temperature monitoring devices. The digital thermometer market is anticipated to outgrow the mercury-based devices in the coming decade.

Considering the market potential and the specific requirements of the country to have indigenous technology for clinical digital thermometers, C-MET has developed a technology for Digital thermometers for medical applications. The thermal sensor probes used are based on an indigenous thermal sensor technology developed by C-MET for various applications.

In our country, digital thermometers for clinical applications are still largely imported. From the recent experience of containing pandemic diseases, our country has realized the need for an indigenous technology for digital thermometers to cater to our needs both in quality and quantity. Hence this technology has been a need of the hour for our health management system.

ABOUT THE PRODUCTS:

The developed clinical digital thermometers are quite handy and can be used to instantly monitor any abnormalities in body temperature. These can find applications in medical clinics, hospitals, and also at homes. Apart from being safer than mercury thermometer, the digital thermometer is provided with a larger digital display than normal, which makes it convenient to instantly monitor the temperature values with better readability.

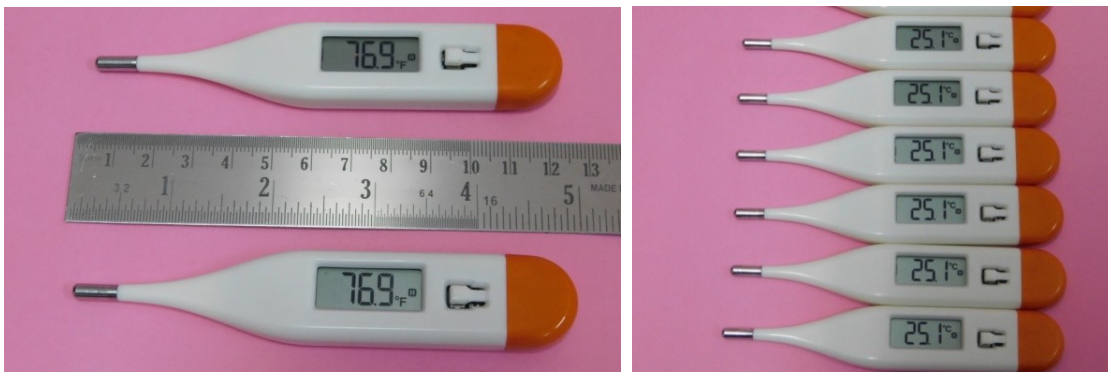


Figure 1: Digital thermometer made at C-MET

Some of the general characteristics of this thermometer are given below.

- Default reading is in Fahrenheit. Can be toggled between degree Fahrenheit and degree Celsius by pressing the switch one time.
- The temperature range of the digital thermometer is 31°C to 43°C. If the temperature is above 43°C it shows 43°C along with "max" indication and if the temperature is below 31°C , it will be 31° C along with "min'' indication
- The resolution of the thermometer is 0.1 ^{o}C and accuracy is ± 0.2 ^{o}C as per Class II of BIS standard
- Once the value goes in hold, "H" Symbol will be shown in the display along with a beep sound
- After every Hold the thermometer goes to sleep automatically in 1 minute 15 seconds.
- "Lo" Indication on display means the battery voltage is low and it is the time to replace the battery
- Large Liquid Crystal Display (LCD) has been provided. The best visibility angle for LCD is 270° or 6'O clock view
- Self checking feature included

The design of the clinical digital thermometer is protected by filing design registration with Indian Patent Office (Design No. 335773-001)

The clinical digital thermometer developed at C-MET has qualified all the tests as per IS 15113:2002 standard for Medical Laboratory instruments – Clinical Electrical Thermometer with Maximum device, conducted by an NABL accredited laboratory.

Technology Transfer

C-MET, Thrissur would like to invite competent industrial partners for commercialization of the above listed product through transfer of technology. The technology will be transferred on non-exclusive basis. The ToT package contains the following

1. Document for design and fabrication
2. Processing details

3. Product specification
4. Test plan and procedure
5. Training

The level of technology readiness at C-MET

The technology readiness level is 6

Period of training under the agreement

C-MET, Thrissur will give 3 weeks training to the engineers of the ToT partner at the premises of C-MET.

Who can apply?

Industries/ start ups can approach C-MET, Thrissur for the technology. Professionally managed Companies and Corporates are also welcome to apply for the technology. A Committee of experts constituted by the C-MET will assess capabilities and strengths of the industry before finalizing the technology partners.

How to apply?

Interested companies may send expression of interest with their details by filling the following questionnaire along with supporting documents to:

Programme Co-ordinator
Centre for Materials for Electronics Technology (C-MET)
Panchawati, Off Pashan Road, Pune:411008
Email: prog.coord@cmet.gov.in

General:

The industry willing to take technology for commercial exploitation will be required to enter into a ToT agreement with C-MET, Thrissur as per the terms and conditions approved by the competent authority in the MeitY prescribed format.

Note:

1. Optionally, for interested parties, Technology on thermal sensor probes (based on one type of chip thermistor) for digital thermometers will be given on additional ToT fee.